

Multiple Treatment Designs

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice working with **multiple treatment experimental designs** and their analyses. The exercises build directly on the concepts introduced in this week's lecture and lab.

You will analyze three agronomic experiments and practice:

- identifying experimental designs
- fitting the correct statistical models
- interpreting main effects and interactions
- visualizing treatment responses

Submit a **knitted PDF generated from your R Markdown file** to Canvas.

Question 1

1 point

Fifteen canola entries were compared in a field trial in Manitoba. The experiment was conducted using a **randomized complete block design with three blocks**. In this dataset block is coded as rep.

Load the dataset "data/canola_data.csv" and print the first six rows.

Question 2

2 points

Fit an appropriate model to evaluate the effect of **canola entry on yield** while accounting for block effects. Print the ANOVA table for your model.

Question 3

2 points

Based on the ANOVA results from Question 2:

- Is there evidence that **canola entry** affects yield?
- Why is it important to include **block** in the model?

Write 2–3 sentences.

Question 4

1 point

Five barley genotypes were evaluated for their response to **three levels of drought stress** in an arid environment. The experiment was conducted as a **factorial treatment structure within a randomized complete block design**.

Load the dataset "data/barley_drought.csv" and print the first six rows.

Question 5

2 points

Fit a factorial model that includes:

- block (you may treat it as fixed or random)
- genotype
- drought
- genotype $\times$ drought interaction

Print the ANOVA table.

Question 6

2 points

Based on the ANOVA table from Question 5, interpret the results.

Address the following questions:

- Is drought stress a significant factor affecting yield?
- Do genotypes differ in yield?
- Is the **genotype:drought interaction** significant?

Question 7

1 point

In your own words, explain what a significant **genotype:drought** interaction means in the context of this experiment.

Question 8

1 point

The effects of **irrigation and tillage on soil nitrite (NO₂ emission)** were studied using a **split-plot design**, where irrigation was applied to main plots and tillage was applied to subplots.

Load the dataset "`data/irrigation_tillage_no2.csv`" and assign it to an object named `nitrite`. Print the first six rows.

Question 9

2 points

Fit an appropriate **split-plot model** for this experiment using a mixed-effects model and print the ANOVA table.

hint: there is a package that starts with `n` and ends with `lme` that fits mixed-effects models.

Question 10

1 point

Based on the ANOVA table and the interaction plot, briefly explain whether irrigation, tillage, or their interaction affects NO₂ emissions.

Question 11 (Bonus)

2 points

This one is optional

In your own words, explain **why split-plot experiments require a different statistical model than completely randomized or randomized complete block designs.**